



ENGINEERING PHYSICS

Course Code:

PH1001-1

Course Type: IPCC

Teaching Hours/Week (L: T: P: S):

3:0:2:0

Credits: 04

Total Teaching Hours:

40+0+26

CIE + SEE Marks: 50+50

Teaching Department: Physics

Course Objectives:

1. To introduce the concepts of wave mechanics to study the properties of sub-atomic particles.
2. To study the concepts of crystalline solids and X-rays.
3. To explain the concepts of semiconductors and semiconductor devices
4. To explain the properties of superconductors and their applications.
5. To explain the principle, working and applications of lasers & optical fibers.

Wave mechanics

UNIT-I

08 Hours

Introduction to wave mechanics. Matter waves – de Broglie's relation, characteristics of matter waves. Wave function, properties and physical significance of a wave function, probability density and normalization of wave function, Schrödinger wave equation (time dependent & independent). Application of Schrödinger wave equation –particle in a potential well of infinite depth, Eigen functions, probability densities and energy Eigen values for a particle in an infinite potential well. Numerical examples.

Crystallography & X-rays

07 Hours

Crystallography: Introduction to crystallography - space lattice, unit cell, primitive cell, lattice parameters. Crystal systems and Bravais lattice. Direction and planes in a crystal, Miller indices – method of finding the Miller indices. Interplanar spacing – derivation. Coordination number, number of atoms per unit cell and atomic packing factor - simple cubic, body centered cubic, and face centered cubic lattices.

X rays: X-rays – generation and properties. Continuous and characteristic X-rays. Bragg's law and Bragg's spectrometer, Applications. Numerical examples.

UNIT-II	
Semiconductors	11 Hours
Semiconductors: Band structure - classification of solids. Semiconductors - intrinsic and extrinsic semiconductors, carrier generation. Direct and indirect band gap semiconductors. Fermi - Dirac Statistics, Fermi factor, Fermi energy level in intrinsic and extrinsic semiconductors and effect of temperature on Fermi level, intrinsic effect - maximum device temperature. Conductivity of intrinsic and extrinsic semiconductors - derivation. Effect of temperature on conductivity of intrinsic and extrinsic semiconductor. Hall effect - derivation of Hall coefficient, carrier concentration and mobility. Applications of Hall effect. Numerical examples. Semiconductor devices: light emitting diode, photodiode, and solar cell.	
Superconductors	04 Hours
Introduction to superconductors, characteristic properties. Type-I and Type-II superconductors. BCS theory (qualitative). Applications of superconductors. Numerical examples.	
UNIT-III	
Lasers	05 Hours
Lasers: Introduction to lasers. Absorption and emission of radiation, Einstein's coefficients. Condition for laser action, population inversion and metastable states. Requisites of a laser system - active medium, pumping mechanism and optical resonant cavity. Three level and four level lasers. Principle, construction and working of Nd:YAG laser, He-Ne laser and semiconductor laser. Applications.	
Optical fibers	05 Hours
Optical fibers: Introduction to optical fibers. Propagation mechanism in optical fibers - angle of acceptance, acceptance cone and numerical aperture - derivation. Fractional index change and V-number. Types of optical fibers and modes of propagation. Attenuation. Applications. Numerical examples.	

Suggested List of Experiments (Any 10 Experiments)

1.	Energy band gap of a semiconductor by four-probe technique.
2.	Hall effect – Determination of the carrier concentration in a semiconductor
3.	Transistor characteristics – Common emitter mode.
4.	Semiconductor laser - Determination of wavelength by diffraction.
5.	Zener diode characteristics – study of current-voltage characteristics
6.	Solar cell – study of its characteristics.
7.	Photo electric effect – Determination of the work function of the material of the emitter of a photocell.
8.	Charging and discharging of a capacitor – Determination of capacitance value, half time and time constant.
9.	Velocity of ultrasonic waves using ultrasonic interferometer
10.	Series and parallel resonance circuits.
11.	LED characteristics.

Course Outcomes: At the end of the course student will be able to

1.	Comprehend various properties of sub-atomic particles on the basis of wave mechanics.
2.	Understand the concepts of crystalline solids, and X-rays.
3.	Understand the concepts of semiconductors and working of semiconductor devices.
4.	Understand the characteristics of superconductors and its applications.
5.	Understand the principle, working and applications of lasers & optical fibers.

Course Outcomes Mapping with Program Outcomes & PSO

Program Outcomes→	1	2	3	4	5	6	7	8	9	10	11	12	PSO↓
↓ Course Outcomes													
PH1001-1.1	3	3											
PH1001-1.2	3	3											
PH1001-1.3	3	3											
PH1001-1.4	3	3											
PH1001-1.5	3	3											

1: Low 2: Medium 3: High

TEXTBOOKS:	
1.	G.K.Shivakumar , Engineering Physics, Prism Engineering Education Series, Prism books Pvt Ltd., Bangalore, 2010-11 edition (Reprint 2013-14).
2.	S. P. Basavaraju , Engineering Physics, Subhas Stores, Bangalore, latest editions.
3.	Arthur Beiser et.al. , Concepts of Modern Physics, Tata McGraaw Hill Education Private Limited, Special Indian Edition, 2009.
REFERENCE BOOKS:	
1.	V. Rajendran , Engineering Physics, Tata McGraw Hill Pub., 2011.
2.	M. R. Srinivasan , Physics for Engineers, New Age International Publishers, Bangalore, 2 nd Edition, 2009.
3.	Kenneth Krane , Modern Physics, Wiley International, 3 rd Edition, 2012.
4.	S. O. Pillai , Solid State Physics, New Age International, 7 th Edition, 2015
5	A.Ghatak , Optics, Tata McGraw Hill Pub., 5 th edition, 2012
6	A. J. Dekker , Electrical Engineering Materials, Prentice Hall India Pub., New Delhi, Reprint 2011.
7	B. G. Streetmann , Solid State Electronic devices, 6 th edition, Prentice Hall India Learning Private Limited.
E Books / MOOCs/ NPTEL	
1.	http://nptel.ac.in/courses/122101002/23
2.	http://nptel.ac.in/courses/113106039/1
3.	http://nptel.ac.in/courses/115106061/